# Type MS · Compression: lossy or lossless?

## Answer 1 of 9 · Question 5 part d · 3 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 5(a) | Award **one** mark for each correct column:<br><br>Translator<br>Characteristic / Assembler / Interpreter / Compiler<br>Does **not** have to be present in memory to execute the translated code. — ✓ — — ✓<br>Translates high level language source code. — — ✓ ✓<br>Executes high level code until either the program has finished, or a syntax error is encountered. — — ✓ —<br>Translates low level language source code into machine code. — ✓ — — | | 3 |
| 5(b) | Award **two** marks for a linked explanation such as:<br><br>• Fewer instructions are executed per second (1) so gameplay will not appear to execute as smoothly / graphics lag / fewer frames per second / can only handle less complex graphics (1)<br><br>• Consoles will execute more instructions per second / have faster processors / GPUs (1) so gameplay will not be realistic / mirrored / fair comparison (1) | One mark can be awarded processing speed implication, but the second mark is dependent on having a gameplay impact being identified.<br>e.g. "A lower clock speed means processing will take longer" is generic for one mark.<br><br>Low performance on its own is not enough. | 2 |

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## Type MS · Compression: lossy or lossless?

### Answer 1 of 9 · Question 5 part d · 3 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

<table>
<tr>
<td>5(c)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Issue</th><th>Utility software</th></tr>
<tr><td>Her word processor cannot read all of the files because it is corrupt.</td><td>File repair/recovery</td></tr>
<tr><td>She has a JPEG image and needs to send it to a publisher who can only read PNG image files.</td><td>Converting</td></tr>
<tr><td>There has been a flood and files on a server have been damaged, but Anna needs to access her data.</td><td>Backup/restore</td></tr>
<tr><td>Anna's hard drive is running very slowly when she tries to access a file.</td><td>Defragmentation/defrag</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>5(d)</td>
<td>Award one mark for:<br>• Lossless (1)<br><br>Award one mark for any of the following up to a maximum of two marks:<br>• No data will be lost / the process is reversable (1)<br>• The manual will still be readable (1)<br>• Removal of data would change the meaning of text (1)<br>• Proofreader will be able to edit the text (1)</td>
<td>Allow converse if candidate gives reasons why lossy would not be used for justification<br><br>Allow RLE as a valid lossless compression method.</td>
<td>3</td>
</tr>
</table>



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## Type MS · Compression: lossy or lossless?

### Answer 1 of 9 · Question 5 part d · 3 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 5(e) | Award **one** mark for each of:<br><br>Method 1<br>• (C AND NOT D) (1)<br>• (NOT C AND D) (1)<br>• <expr1> OR <expr2> (1)<br>Method 2<br>• (C OR D) (1)<br>• NOT (C AND D) (1)<br>• <expr1> AND <expr2> (1)<br><br>Example statements:<br>(C AND NOT D) OR (NOT C AND D)<br>(C OR D) AND NOT (C AND D) | Order of precedence is NOT, AND, OR<br><br>Note other equivalent solutions<br><br>MAX 2 if all terms correct apart from missing brackets that are required.<br><br>Do not credit circuit diagrams. | 3 |
| | | **Total for question 5** | **15** |

## Type MS · Compression: lossy or lossless?

### Answer 1 of 9 · Question 5 part d · 3 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Compression: lossy or lossless?

Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part c.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a) | Any **two** from:<br><br>Kilobyte is a SI (physics/science) / International System of Units measurement, whereas Kibibyte is an IEC (computing) / International Electrotechnical Commission / International System of Quantities / ISO measurement (1)<br><br>Kilobyte is equivalent to 1000 bytes, whereas a kibibyte is equivalent to 1024 bytes (1)<br><br>Kilobyte is equivalent to $10^3$ bytes, whereas a kibibyte is equivalent to $2^{10}$ bytes (1)<br><br>Kilobyte is a base 10 measurement, whereas a kibibyte is a base 2 measurement (1) | Ignore units | 2 |

## Type MS · Compression: lossy or lossless?

### Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(b) | Award **4 marks** for 19.17<br><br>or<br><br>**One** mark for each of:<br><br>• Calculates correct bits/bytes(1)<br>• Adds metadata (1)<br>• Calculates correct megabytes (1)<br>• Completely correct response (1)<br><br>Examples:<br><br>$\frac{\frac{2322 \times 4128 \times 16}{8} + 975}{1000 \times 1000}$<br><br>$\frac{(2322 \times 4128 \times 2) + 975}{1000 \times 1000}$<br><br>((2322 x 4128 x 16 bits per colour) ÷ 8 bits per byte) + 975 bytes<br>---<br>1000 bytes per KB x 1000 kilobytes per MB<br><br>((2322 x 4128 x 2 bytes per colour) + 975 bytes<br>---<br>1000 bytes per KB x 1000 kilobytes per MB | • Arithmetic precedence rules must be clearly followed<br>• Units are not required<br>• Equivalent expressions are accepted<br>• Calculations not explicit (8x1000x1000) but expressed (8000000) gain the mark | 4 |

## Type MS · Compression: lossy or lossless?

### Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(i) | Any **two** from:<br>• To save disc/storage space (1)<br>• Reduce transmission time (1)<br>• Meet email attachment size restrictions (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(ii) | Any **two** from:<br>• Compressing / decompressing the file takes execution time (1)<br>• Both compressing and decompressing have to be done with compatible software / both ends of transmission have to use compatible software (1)<br>• (If the compression utility uses a lossy algorithm then) some of the data will be permanently removed / it won't be possible to reconstruct the original file. (1)<br>• Can affect the quality of the file (1)<br>• File size can increase (1) | | 2 |

## Type MS · Compression: lossy or lossless?

### Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(d) | Award 4 marks for 11<br>Award 3 marks for a value between 10 and 11<br><br>**One** mark for each:<br><br>• 80 x 66 x 2 or 80 x 66 x 16 ÷ 8 (1)<br>• + 40 or + 80 (1)<br>• ÷ 1024 (1)<br>• Ceiling or RoundUp (1) | • Arithmetic precedence rules must be clearly followed<br>• Units are not required<br>• Alternative notations for ceiling are accepted<br>• If 16 bits per Unicode character is not awarded, then allow follow through error<br>• Allow follow through error, if interim calculations attempted<br>• Equivalent expressions are accepted | 4 |

## Type MS · Compression: lossy or lossless?

### Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 5 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Mark |
|---|---|---|

Type MS · Compression: lossy or lossless?

Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part c.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a) | Any **two** from:<br><br>Kilobyte is a SI (physics/science) / International System of Units measurement, whereas Kibibyte is an IEC (computing) / International Electrotechnical Commission / International System of Quantities / ISO measurement (1)<br><br>Kilobyte is equivalent to 1000 bytes, whereas a kibibyte is equivalent to 1024 bytes (1)<br><br>Kilobyte is equivalent to $10^3$ bytes, whereas a kibibyte is equivalent to $2^{10}$ bytes (1)<br><br>Kilobyte is a base 10 measurement, whereas a kibibyte is a base 2 measurement (1) | Ignore units | 2 |

## Type MS · Compression: lossy or lossless?

### Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(b) | Award **4 marks** for 19.17<br><br>or<br><br>**One** mark for each of:<br><br>• Calculates correct bits/bytes(1)<br>• Adds metadata (1)<br>• Calculates correct megabytes (1)<br>• Completely correct response (1)<br><br>Examples:<br><br>$\frac{\frac{2322 \times 4128 \times 16}{8} + 975}{1000 \times 1000}$<br><br>$\frac{(2322 \times 4128 \times 2) + 975}{1000 \times 1000}$<br><br>((2322 x 4128 x 16 bits per colour) ÷ 8 bits per byte) + 975 bytes<br>---<br>1000 bytes per KB x 1000 kilobytes per MB<br><br>((2322 x 4128 x 2 bytes per colour) + 975 bytes<br>---<br>1000 bytes per KB x 1000 kilobytes per MB | • Arithmetic precedence rules must be clearly followed<br>• Units are not required<br>• Equivalent expressions are accepted<br>• Calculations not explicit (8x1000x1000) but expressed (8000000) gain the mark | 4 |

## Type MS · Compression: lossy or lossless?

### Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(i) | Any **two** from:<br>• To save disc/storage space (1)<br>• Reduce transmission time (1)<br>• Meet email attachment size restrictions (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(ii) | Any **two** from:<br>• Compressing / decompressing the file takes execution time (1)<br>• Both compressing and decompressing have to be done with compatible software / both ends of transmission have to use compatible software (1)<br>• (If the compression utility uses a lossy algorithm then) some of the data will be permanently removed / it won't be possible to reconstruct the original file. (1)<br>• Can affect the quality of the file (1)<br>• File size can increase (1) | | 2 |

## Type MS · Compression: lossy or lossless?

### Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(d) | Award 4 marks for 11<br>Award 3 marks for a value between 10 and 11<br><br>**One** mark for each:<br><br>• 80 x 66 x 2 or 80 x 66 x 16 ÷ 8 (1)<br>• + 40 or + 80 (1)<br>• ÷ 1024 (1)<br>• Ceiling or RoundUp (1) | • Arithmetic precedence rules must be clearly followed<br>• Units are not required<br>• Alternative notations for ceiling are accepted<br>• If 16 bits per Unicode character is not awarded, then allow follow through error<br>• Allow follow through error, if interim calculations attempted<br>• Equivalent expressions are accepted | 4 |

## Type MS · Compression: lossy or lossless?

### Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 5 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Mark |
|---|---|---|

Type MS · Compression: lossy or lossless?

Answer 4 of 9 · Question 4 part c · 1 mark

2020 NOV · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a)(i) | Digital (signal) / converted analogue (signal) (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(a)(ii) | Analogue (signal) / analogue (sound wave) (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(a)(iii) | 2 Hertz / hertz /Hz (1) | • Do not penalise spelling<br>• Accept 2 samples/cycles per second | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(a)(iv) | Any **one** mark for:<br>• The (analogue sound) wave will be represented more accurately / the fidelity/quality of the recording is improved (1)<br>• The file size will increase / more data stored (as each sample takes up disk space) (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(a)(v) | • 3 (1)<br>• The range of sampled sound is from 0-6. To store 7 distinct values, you need 3 bits (111 binary = 7 denary) (1)<br>OR graph shows 8 possible values, 3 bits can store 0 - 7 / 8 values (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

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## Type MS · Compression: lossy or lossless?

### Answer 4 of 9 · Question 4 part c · 1 mark

2020 NOV · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a)(vi) | Correct binary value 101 (1) | Ignore leading zeros | 1 |



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## Type MS · Compression: lossy or lossless?

### Answer 4 of 9 · Question 4 part c · 1 mark

2020 NOV · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(b) | 819 seconds * 8 bytes * 47,000 hertz<br>819 x 8 x 47000<br>819 seconds * 64÷8 bytes * 47,000 hertz<br>818 x 64 x 47000<br>8<br><br>819 (1)<br>8 or 64÷8 (1)<br>47,000 (1)<br>Fully correct expression (1) | Award equivalent expressions | 4 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c) | Any **one** from:<br>• The sound quality may be poorer/lower (for some people) (1)<br>• Cannot get the original back after compression (1) | | 1 |



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## Type MS · Compression: lossy or lossless?

### Answer 4 of 9 · Question 4 part c · 1 mark

2020 NOV · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Compression: lossy or lossless?

### Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 1 of 5 · find part f.i in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 2(a) | Award **one** mark from:<br><br>• share peripherals/devices (e.g. printers, scanners) (1)<br>• communicate (e.g. email, instant messaging, play games) (1)<br>• share data (e.g. files/music/videos/backups on servers) (1)<br>• deploy/update applications/software(1)<br>• administrative purposes (e.g. remote access/desktop) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | Award **one** mark from:<br><br>• to allow (connected) machines to communicate (1)<br>• to provide the rules of communication between (networked) devices (1) | Accept any other response indicating protocols enable communication between devices | 1 |

## Type MS · Compression: lossy or lossless?

### Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 2 of 5 · find part f.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | **The only correct answer is A**<br><br>***B*** *is not correct because FTP is an application protocol, which is added in the application layer*<br><br>***C*** *is not correct because FTP is an application protocol, which is added in the application layer*<br><br>***D*** *is not correct because FTP is an application protocol, which is added in the application layer* | | 1 |
| 2(c)(ii) | Award **one** mark from:<br><br>• Adds the source/destination IP address (1)<br>• Establishes sockets (an IP address followed by a port number) (1)<br>• Moves packets onto the next network node (1)<br>• Strips the source/destination IP addresses (when it arrives at its destination) (1)<br>• Passes the packet to the transport/next layer (when it arrives at its destination) (1)<br>• Adds/removes packet headers (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d) | Award **one** mark for each of:<br><br>**Domain Name** / **IP Address** / **URL**<br>(blank) / (blank) / ✓<br>(blank) / ✓ / (blank)<br>✓ / (blank) / (blank) | | 3 |

# Type MS · Compression: lossy or lossless?

## Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 3 of 5 · find part f.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(e)(i) | Bus | | 1 |
| 2(e)(ii) | Award **one** mark from:<br>• Easy to connect/remove devices/computers (in a network) (1)<br>• Cable cost less / fewer cables needed (compared to mesh/star topology) (1)<br>• Easier to plan/setup/maintain (1) | | 1 |
| 2(e)(iii) | Award **one** mark from:<br>• If the main cable/a terminator fails or gets damaged the whole network will fail (1)<br>• The more workstations the slower it gets (1)<br>• The more workstations the more collisions (1)<br>• Every device 'sees' all of the data on the network (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(f)(i) | Award **two** marks from:<br>• Reduces time taken to upload/download (1)<br>• Reduces the bandwidth required/cost of data to upload/download (1)<br>• (Usually) produces smaller file/reduces storage (on the web server/device) (1)<br>• People have a limited hearing range / can't tell the difference (1)<br>• People's brains will 'fill in the blanks' (1) | | 2 |
| 2(f)(ii) | **The only correct answer is C**<br><br>***A*** *is not correct because it is not compressed*<br><br>***B*** *is not correct because it uses lossless compression*<br><br>***D*** *is not correct because it uses lossless compression* | | 1 |
| 2(f)(iii) | Award **one** mark from:<br>• 3c3w2c6w3c (1)<br>• c3w3c2w6c3 (1) | | 1 |

## Type MS · Compression: lossy or lossless?

### Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 4 of 5 · find part f.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(g) | Award **four** marks for a correct expression or correct answer<br><br>**Examples of correct expressions:**<br>6 x 1000 x 1000 x 1000 x 8 (4)<br>54 x 1000 x 1000<br><br>1000 x 8 (4)<br>9<br><br>888.9 (4)<br><br>For an incorrect or partial expression award **one** mark for each of:<br><br>• **Bits to transfer**<br>○ $6 \times 1000^3$ (1)<br>○ x 8 (or equivalent) (1)<br><br>• **Speed in bits per second**<br>○ $54 \times 1000^2$ (1)<br><br>• Their bits divided by their speed (1) | • Units are not required<br>• Equivalent expressions are accepted<br>• Allow follow through for incorrect or partial expressions | 4 |

## Type MS · Compression: lossy or lossless?

### Answer 5 of 9 · Question 2 part f.i · 2 marks

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Compression: lossy or lossless?

Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 5 · find part f.ii in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 2(a) | Award **one** mark from:<br><br>• share peripherals/devices (e.g. printers, scanners) (1)<br>• communicate (e.g. email, instant messaging, play games) (1)<br>• share data (e.g. files/music/videos/backups on servers) (1)<br>• deploy/update applications/software(1)<br>• administrative purposes (e.g. remote access/desktop) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | Award **one** mark from:<br><br>• to allow (connected) machines to communicate (1)<br>• to provide the rules of communication between (networked) devices (1) | Accept any other response indicating protocols enable communication between devices | 1 |

## Type MS · Compression: lossy or lossless?

### Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part f.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | **The only correct answer is A**<br><br>***B*** *is not correct because FTP is an application protocol, which is added in the application layer*<br><br>***C*** *is not correct because FTP is an application protocol, which is added in the application layer*<br><br>***D*** *is not correct because FTP is an application protocol, which is added in the application layer* | | 1 |
| 2(c)(ii) | Award **one** mark from:<br><br>• Adds the source/destination IP address (1)<br>• Establishes sockets (an IP address followed by a port number) (1)<br>• Moves packets onto the next network node (1)<br>• Strips the source/destination IP addresses (when it arrives at its destination) (1)<br>• Passes the packet to the transport/next layer (when it arrives at its destination) (1)<br>• Adds/removes packet headers (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d) | Award **one** mark for each of:<br><br>**Domain Name** / **IP Address** / **URL**<br>(blank) / (blank) / ✓<br>(blank) / ✓ / (blank)<br>✓ / (blank) / (blank) | | 3 |

## Type MS · Compression: lossy or lossless?

### Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 3 of 5 · find part f.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(e)(i) | Bus | | 1 |
| 2(e)(ii) | Award **one** mark from:<br>• Easy to connect/remove devices/computers (in a network) (1)<br>• Cable cost less / fewer cables needed (compared to mesh/star topology) (1)<br>• Easier to plan/setup/maintain (1) | | 1 |
| 2(e)(iii) | Award **one** mark from:<br>• If the main cable/a terminator fails or gets damaged the whole network will fail (1)<br>• The more workstations the slower it gets (1)<br>• The more workstations the more collisions (1)<br>• Every device 'sees' all of the data on the network (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(f)(i) | Award **two** marks from:<br>• Reduces time taken to upload/download (1)<br>• Reduces the bandwidth required/cost of data to upload/download (1)<br>• (Usually) produces smaller file/reduces storage (on the web server/device) (1)<br>• People have a limited hearing range / can't tell the difference (1)<br>• People's brains will 'fill in the blanks' (1) | | 2 |
| 2(f)(ii) | **The only correct answer is C**<br><br>***A*** *is not correct because it is not compressed*<br><br>***B*** *is not correct because it uses lossless compression*<br><br>***D*** *is not correct because it uses lossless compression* | | 1 |
| 2(f)(iii) | Award **one** mark from:<br>• 3c3w2c6w3c (1)<br>• c3w3c2w6c3 (1) | | 1 |

## Type MS · Compression: lossy or lossless?

### Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part f.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(g) | Award **four** marks for a correct expression or correct answer<br><br>**Examples of correct expressions:**<br>6 x 1000 x 1000 x 1000 x 8 (4)<br>54 x 1000 x 1000<br><br>1000 x 8 (4)<br>9<br><br>888.9 (4)<br><br>For an incorrect or partial expression award **one** mark for each of:<br><br>• **Bits to transfer**<br>○ $6 \times 1000^3$ (1)<br>○ x 8 (or equivalent) (1)<br><br>• **Speed in bits per second**<br>○ $54 \times 1000^2$ (1)<br><br>• Their bits divided by their speed (1) | • Units are not required<br>• Equivalent expressions are accepted<br>• Allow follow through for incorrect or partial expressions | 4 |

## Type MS · Compression: lossy or lossless?

### Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 5 of 5 · find part f.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Compression: lossy or lossless?

### Answer 7 of 9 · Question 1 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part b.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(ai) | 0100 1101<br>100 1101<br><br>Award **one** mark for each correct nibble | Must be in the correct order | 2 |
| 1(a)(ii) | 1000 0010<br><br>Award **one** mark for each correct nibble | Must be in the correct order | 2 |
| 1(a)(iii) | Award **one** mark for each of:<br><br>• C (1)<br>• 6 (1) | Must be in the correct order | 2 |

## Type MS · Compression: lossy or lossless?

### Answer 7 of 9 · Question 1 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part b.iii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b)(i) | 400 pixels x 200 pixels x 12 bits / 8 bits per byte<br>Allow expression in word form<br><br>$\frac{400 \times 200 \times 12}{8}$<br><br>400 x 200 **OR** 80000(1)<br><br>12/8 **OR** 1.5 (1) | Accept any other use of 400, 200, 12, 8 that gives the correct answer<br><br>Allow one mark for 120,000 with no expression | 2 |
| 1(b)(ii) | Spaces are to help legibility.<br>Award **three** marks for:<br><br>Line 1 3W 1B 4W<br>Line 2 1W 6B 1W<br>Line 3 1B 2W 1B 1W 1B 2W<br>Line 4 3W 2B 3W<br><br>Award **two** marks for any three lines correctly encoded<br><br>Award **one** mark for any two lines correctly encoded | Allow **one** error in the sequence<br><br>Accept letter before number but must be consistently used | 3 |
| 1(b)(iii) | Award **one** mark from:<br><br>• None of the original (image) detail/quality is lost (when the image is stored) (1)<br>• None of the data/information is (permanently) removed (1)<br>• It can be decompressed without losing detail (1) | | 1 |

## Type MS · Compression: lossy or lossless?

### Answer 7 of 9 · Question 1 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part b.iii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c)(i) | Award up to **three** marks for a linked description:<br>• set the sample rate/parameters/bit-depth (1)<br>• sample (the analogue sound) (1)<br>• measure the sound amplitude/volume/frequency (1)<br>• give a (binary) value/number for each measurement (1)<br>• store data as sample rate and values / digital signals (1) | | 3 |
| 1(c)(ii) | Award **one** mark from:<br>• sound may be missing some frequencies (1)<br>• some audio information/data is lost/deleted during the (compression) process (1) | | 1 |
| | | **Total** | **16** |

## Type MS · Compression: lossy or lossless?

### Answer 7 of 9 · Question 1 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 4 of 4 · find part b.iii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Compression: lossy or lossless?

### Answer 8 of 9 · Question 1 part c.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(ai) | 0100 1101<br>100 1101<br><br>Award **one** mark for each correct nibble | Must be in the correct order | 2 |
| 1(a)(ii) | 1000 0010<br><br>Award **one** mark for each correct nibble | Must be in the correct order | 2 |
| 1(a)(iii) | Award **one** mark for each of:<br><br>• C (1)<br>• 6 (1) | Must be in the correct order | 2 |

## Type MS · Compression: lossy or lossless?

### Answer 8 of 9 · Question 1 part c.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b)(i) | 400 pixels x 200 pixels x 12 bits / 8 bits per byte<br>Allow expression in word form<br><br>$\frac{400 \times 200 \times 12}{8}$<br><br>400 x 200 **OR** 80000(1)<br><br>12/8 **OR** 1.5 (1) | Accept any other use of 400, 200, 12, 8 that gives the correct answer<br><br>Allow one mark for 120,000 with no expression | 2 |
| 1(b)(ii) | Spaces are to help legibility.<br>Award **three** marks for:<br><br>Line 1 3W 1B 4W<br>Line 2 1W 6B 1W<br>Line 3 1B 2W 1B 1W 1B 2W<br>Line 4 3W 2B 3W<br><br>Award **two** marks for any three lines correctly encoded<br><br>Award **one** mark for any two lines correctly encoded | Allow **one** error in the sequence<br><br>Accept letter before number but must be consistently used | 3 |
| 1(b)(iii) | Award **one** mark from:<br><br>• None of the original (image) detail/quality is lost (when the image is stored) (1)<br>• None of the data/information is (permanently) removed (1)<br>• It can be decompressed without losing detail (1) | | 1 |

## Type MS · Compression: lossy or lossless?

### Answer 8 of 9 · Question 1 part c.ii · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c)(i) | Award up to **three** marks for a linked description:<br>• set the sample rate/parameters/bit-depth (1)<br>• sample (the analogue sound) (1)<br>• measure the sound amplitude/volume/frequency (1)<br>• give a (binary) value/number for each measurement (1)<br>• store data as sample rate and values / digital signals (1) | | 3 |
| 1(c)(ii) | Award **one** mark from:<br>• sound may be missing some frequencies (1)<br>• some audio information/data is lost/deleted during the (compression) process (1) | | 1 |
| | | **Total** | **16** |

## Type MS · Compression: lossy or lossless?

### Answer 8 of 9 · Question 1 part c.ii · 1 mark

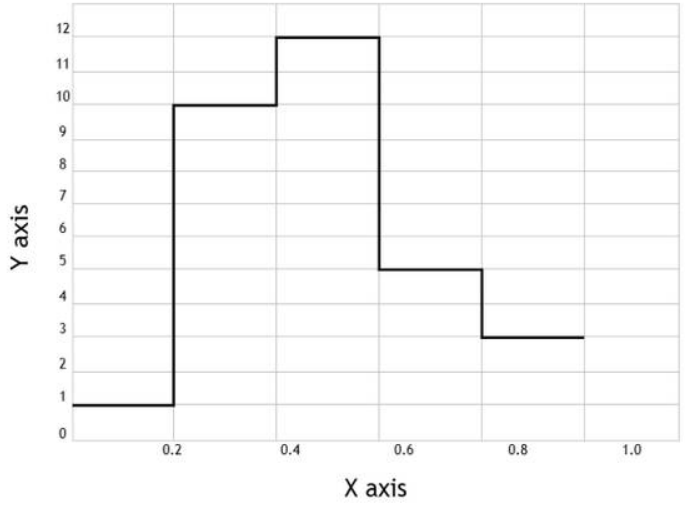
2021 NOV · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Compression: lossy or lossless?

Answer 9 of 9 · Question 3 part b.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:<br><br>• At least four conversions plotted at the correct amplitude (1)<br>• Correct start point (1) 0 – 1 digital. 0 – 0 analogue<br>• Digital sound wave drawn (1) using candidate's plots<br><br><table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table><br> 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:<br><br>• Time (1)<br>• Sample interval/period (1)<br>• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:<br><br>• Amplitude / sound level / volume (1)<br>• Metres/centimetres/nanometres (1)<br>• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

## Type MS · Compression: lossy or lossless?

### Answer 9 of 9 · Question 3 part b.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:<ul><li>Transfer time would be longer (1)</li><li>Larger file size / less compression (1)</li><li>Takes up more storage space (1)</li><li>Uses more of her data allocation (1)</li></ul></td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:<ul><li>Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)</li><li>Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)</li><li>Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)</li><li>Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</li></ul></td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:<ul><li>Storage host could be targeted by hackers (1)</li><li>Alyssa has less control over her files (1)</li><li>An untrustworthy employee (of the cloud storage provider) could steal her files (1)</li><li>Reliant on the storage provider for security / safekeeping (1)</li><li>Files could be intercepted/corrupted during upload/download (1)</li></ul></td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · Compression: lossy or lossless?

Answer 9 of 9 · Question 3 part b.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | ru2.mp3 | File/media/resource wanted (on the website)/ path / part of path (1) | | |
| | | | **Total for question 3** | **13** |

## Type MS · Compression: lossy or lossless?

### Answer 9 of 9 · Question 3 part b.i · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

| Question Number | Answer (flow chart replaced) | Additional Guidance | Mark |
|---|---|---|---|